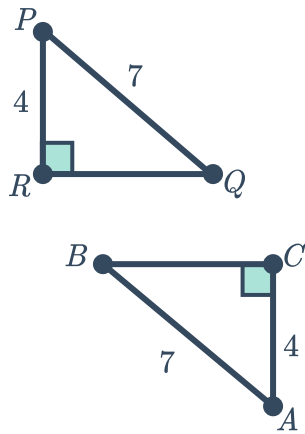


Triangle congruence tests

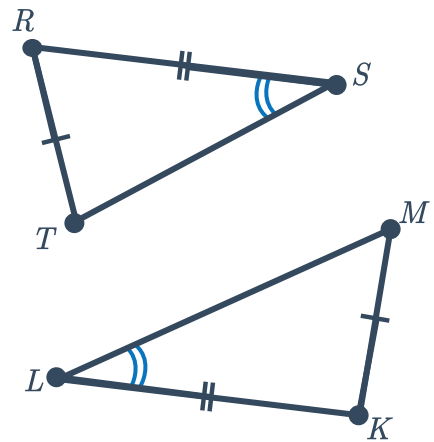


- 1 For each pair of triangles, state whether the two triangles are congruent. If yes, state what congruence test they satisfy.

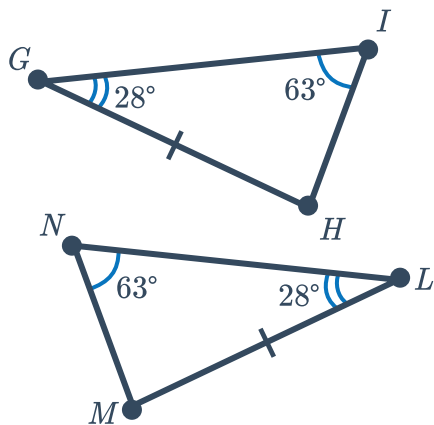
a



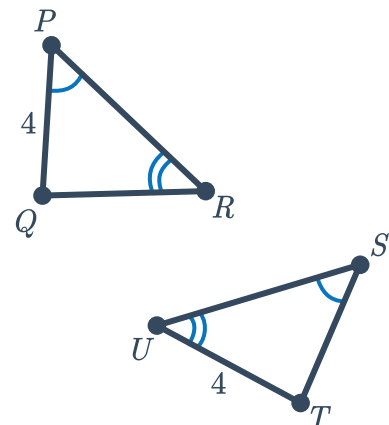
b



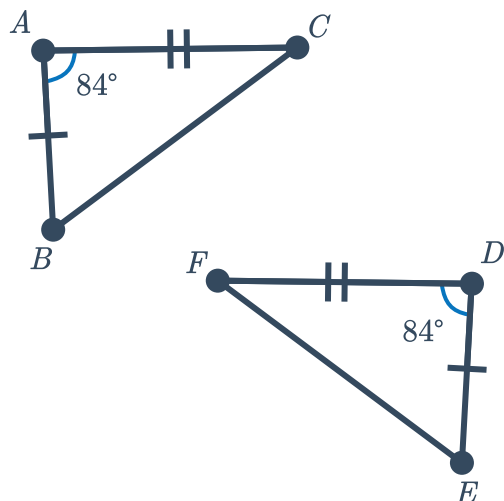
c



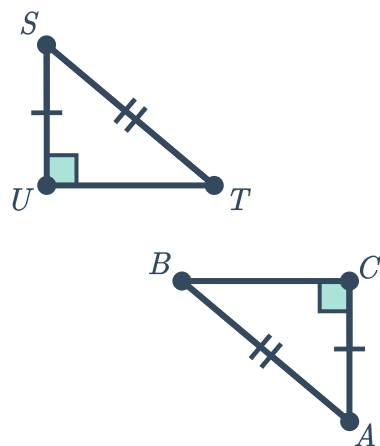
d



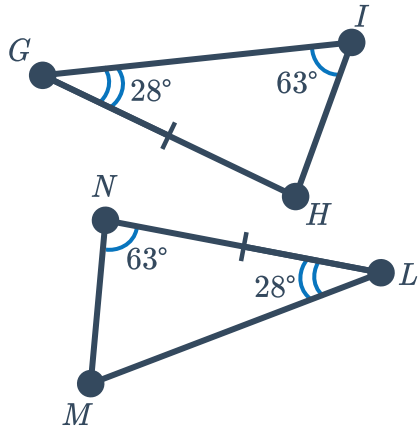
e



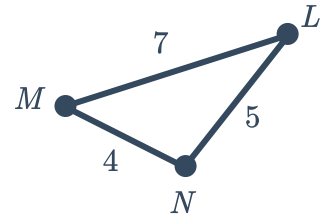
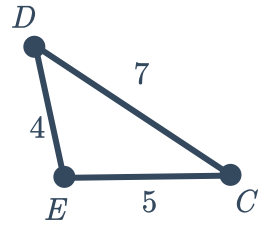
f



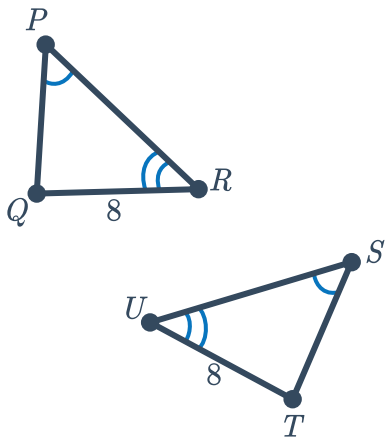
g



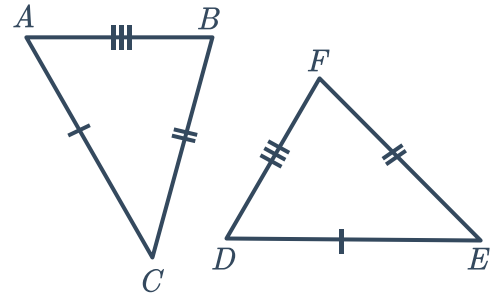
h



i

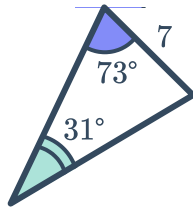


j

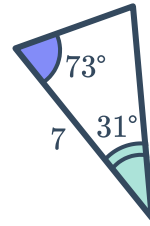


2 Consider the following triangles:

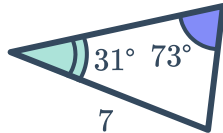
Triangle A



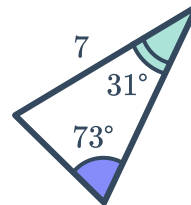
Triangle B



Triangle C



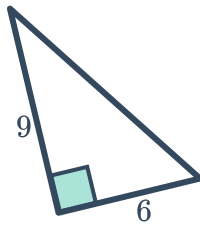
Triangle D



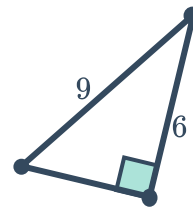
- a State which two triangles are congruent.
- b State the congruence test this pair satisfies.

3 Consider the following triangles:

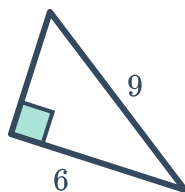
Triangle A



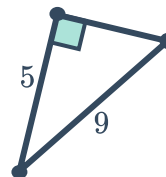
Triangle B



Triangle C



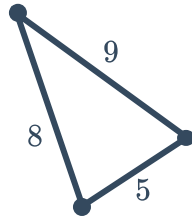
Triangle D



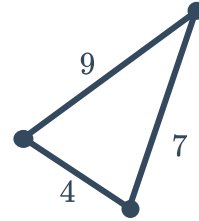
- a State which two triangles are congruent.
- b State the congruence test that this pair satisfies.

4 Consider the following triangles:

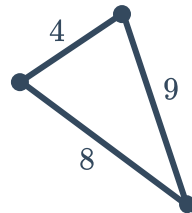
Triangle A



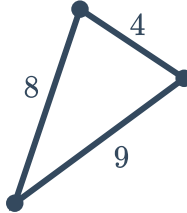
Triangle B



Triangle C



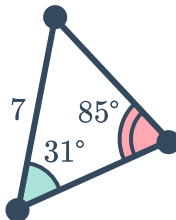
Triangle D



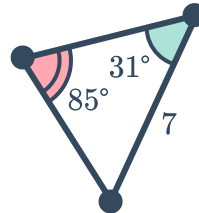
- a State which two triangles are congruent.
- b State the congruence test that this pair satisfies.

5 Consider the following triangles:

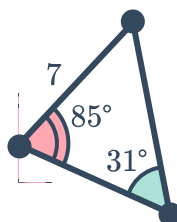
Triangle A



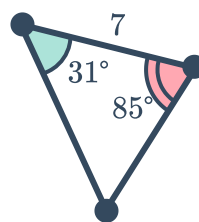
Triangle B



Triangle C



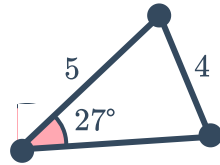
Triangle D



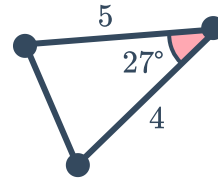
- a State which two triangles are congruent.
- b State the congruence test that this pair satisfies.

6 Consider the following triangles:

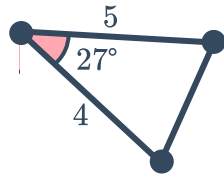
Triangle A



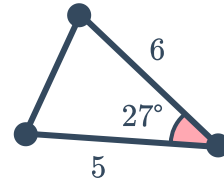
Triangle B



Triangle C



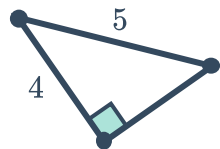
Triangle D



- a State which two triangles are congruent.
- b State the congruence test that this pair satisfies.

7 Consider the following triangles:

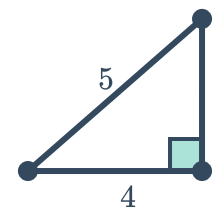
Triangle A



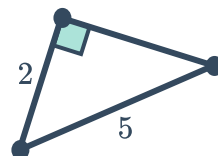
Triangle B



Triangle C



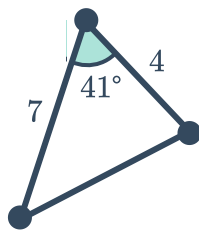
Triangle D



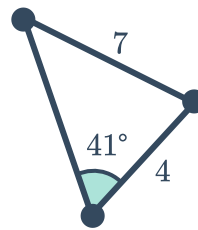
- a State which two triangles are congruent.
- b State the congruence test that this pair satisfies.

8 Consider the following triangles:

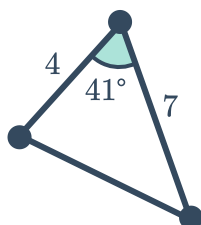
Triangle A



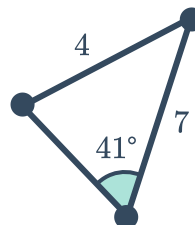
Triangle B



Triangle C



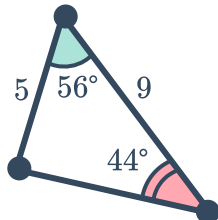
Triangle D



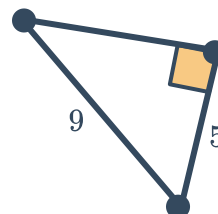
- a State which two triangles are congruent.
- b State the congruence test that this pair satisfies.

9 Consider the following triangles:

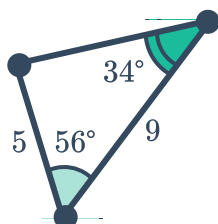
Triangle A



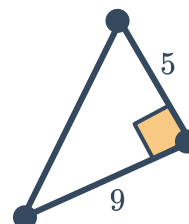
Triangle B



Triangle C



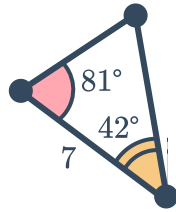
Triangle D



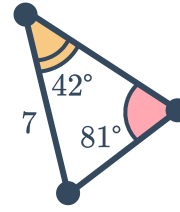
- a State which two triangles are congruent.
- b State the congruence test that this pair satisfies.

10 Consider the following triangles:

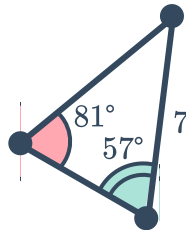
Triangle A



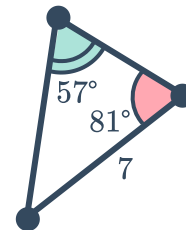
Triangle B



Triangle C

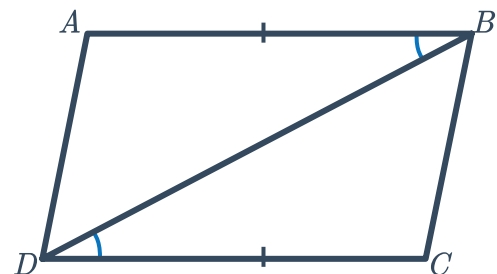


Triangle D



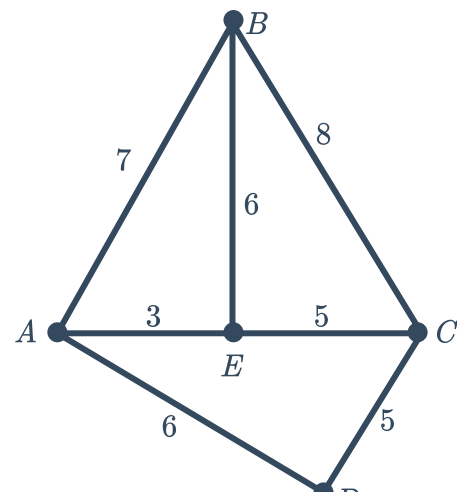
- a State which two triangles are congruent.
- b State the congruence test that this pair satisfies.

11 Consider the given diagram:
State the congruence test that triangles $\triangle ABD$ and $\triangle CDB$ satisfy.



12 Consider the given diagram:

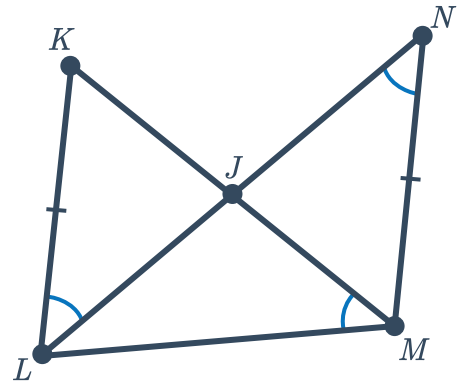
- a State which two triangles are congruent.
- b State the congruence test that this pair satisfies.





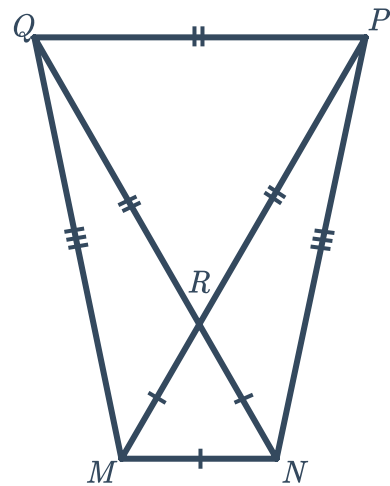
13 Consider the given diagram where KM and LN are straight line segments:

- a State which two triangles are congruent.
- b State the congruence test that this pair satisfies.



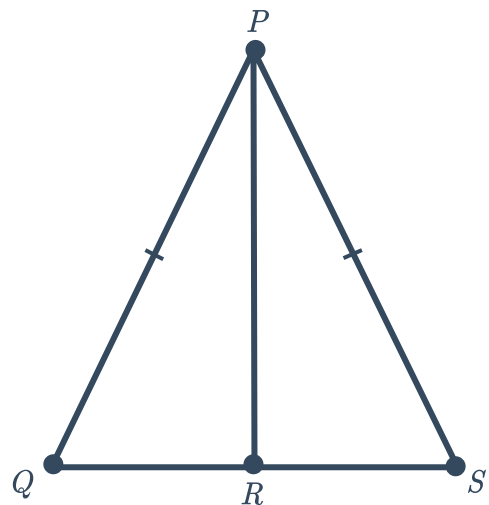
14 Consider the given diagram:

- a State which two triangles that are congruent.
- b State the congruence test that this pair satisfies.



15 Consider the given diagram:

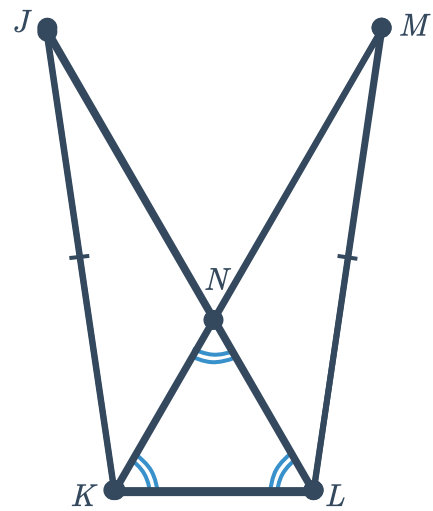
What information is needed so that the triangles $\triangle PQR$ and $\triangle PSR$ are congruent?



16 Consider the given diagram:

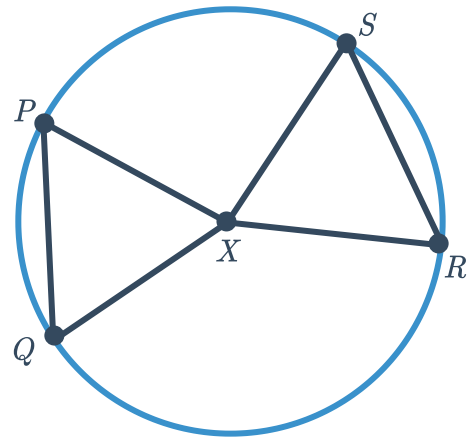


What information is needed so that the triangles $\triangle JKN$ and $\triangle MLN$ are congruent?



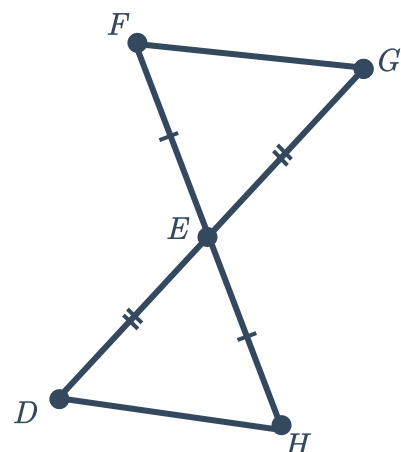
17 Consider the given diagram where X is the centre of the circle.

What information is needed so that the triangles $\triangle PQX$ and $\triangle RSX$ are congruent?



18 Consider the given diagram:

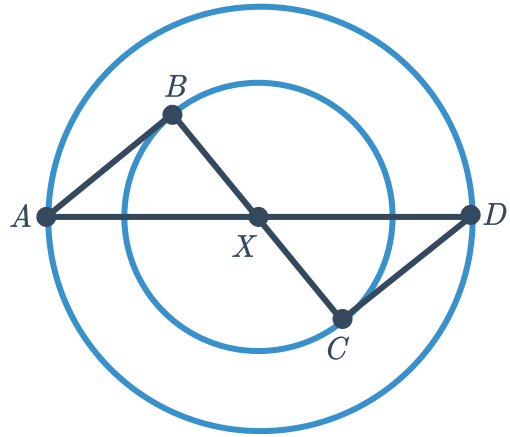
- a State the congruence test that triangles $\triangle EFG$ and $\triangle EHD$ satisfy.
- b State the three ways in which the triangles satisfy this congruence test.



- 19 Consider the given diagram where X is the centre of both circles:

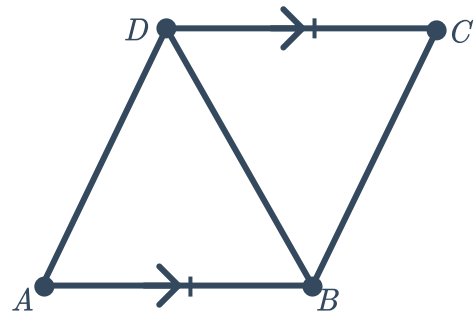


- a State the congruence test that triangles $\triangle ABX$ and $\triangle DCX$ satisfy.
- b State the three ways in which the triangles satisfy this congruence test.



- 20 Consider the given diagram:

- a State the congruence test that triangles $\triangle ABD$ and $\triangle CDB$ satisfy.
- b State the three ways in which the triangles satisfy this congruence test.



- 21 Consider the given diagram:

- a State the congruence test that triangles $\triangle ABX$ and $\triangle DCX$ satisfy.
- b State the three ways in which the triangles satisfy this congruence test.

